

BALALAIKA-LONGFORM: A RUSSIAN SPEECH CORPUS FOR CONTINUOUS LONG-FORM TEXT-TO-SPEECH

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ABSTRACT

Long-form text-to-speech must retain requested words over extended generations, yet sentence-level training and evaluation can hide omissions and early stops. We introduce Balalaika-Longform, an openly released Russian corpus of 189 hours in continuous units of 30 seconds to 15 minutes. Long units and matched short windows derived from them support fine-tuning comparisons on the same source recordings, and the accompanying evaluation retains every synthesis attempt. We fine-tune CosyVoice3, Qwen3-TTS, VoxCPM2 and F5-TTS on either view of the corpus and test continuous synthesis on 50 fixed text-voice pairs with voices unseen in fine-tuning, at about 75, 300 and 1,200 input words. At approximately 1,200 input words, CosyVoice3 WER falls from 99.9% after short-window fine-tuning to 47.3% after long-target fine-tuning, and to 16.6% with punctuated, format-matched transcripts; paired intervals also favor long targets for Qwen3-TTS and VoxCPM2, and by a small margin for F5-TTS, whose absolute WER stays above 90%. These results characterize the effect of preserving training sequence length under a fixed continuous-generation protocol.

Index Terms— text-to-speech, Russian speech, speech corpus, long-form synthesis

1. INTRODUCTION

Audiobooks, lectures and podcasts require speech synthesis to preserve content and voice beyond a sentence. Short outputs can conceal omissions, repetitions and early stops, and listeners judge sentences and paragraphs differently [1]. Existing pipelines obtain long audio by splitting text into short generations [2, 3], but splitting discards cross-sentence prosodic and discourse context and cannot test whether a model can sustain a reading by itself. Studying that capability requires data and evaluation that retain long context. Chunked synthesis is a different operating regime and outside our scope: we study how models behave when a long text is given in a single pass.

We introduce Balalaika-Longform for this purpose in Russian. The Balalaika pipeline [4] supplies context-preserving segmentation, transcripts, provenance and quality screening; the corpus retains continuous speech units up to 15 minutes. Paired short-window and long-sequence training views allow a source-controlled comparison of sequence length, with channel-disjoint train, development and test splits.

We evaluate these views on CosyVoice3, Qwen3-TTS, VoxCPM2 and F5-TTS. A controlled two-voice CosyVoice3 experiment tests duration under matched training exposure (Table 2); a published-text comparison of 50 fixed text-voice pairs from 22 works (Table 3) retains every failure in word-error and correct-word-recall estimates, with paired Short/Long intervals for all four backbones.

Our contributions are the long-form corpus, paired training views and an all-attempt evaluation of corpus use across generation families. The experiments separate this data question from the choice to generate continuously or in chunks; transcript-source, formatting and corpus-composition confounds are stated explicitly.

2. RELATED WORK

Speech corpora for synthesis. Open TTS corpora make data quality and coverage reproducible research choices. LibriTTS-R improves the acoustic quality of an existing English corpus through speech restoration [5], HiFiTTS-2 scales audiobook-derived data with quality filtering and detailed metadata [6], and Libriheavy keeps punctuation, casing and preceding context for 50,000 hours of audiobook speech [7]. For Russian, RUSLAN provides carefully recorded single-speaker speech [8], and Balalaika provides a pipeline for context-preserving segmentation, multi-ASR transcription, and speaker-purity screening [4]. We build on the latter to produce a corpus organized around long continuous speech units with paired short and long training views.

Approaches to long-form generation. Recent work addresses long speech through inference and model design. MagpieTTS-LF carries state and text history across sentence chunks without retraining [9]. VoiceStar combines positional modeling and a training scheme for duration control and extrapolation [10]. SpeechSSM samples minute speech and introduces the LibriSpeech-Long benchmark for English [11]. F5-TTS offers a non-autoregressive flow-matching alternative [12]; its standard long-text inference splits the text and joins separately synthesized chunks [3]. Our study instead examines the contribution of training data: whether preserving long targets helps existing models under a fixed uninterrupted-inference protocol, without assuming that long-form training is necessary for every synthesis method.

Backbones and evaluation. CosyVoice [13, 14], Qwen3-TTS [15], and VoxCPM2 [16] provide open autoregressive models with different speech representations; we use their public fine-tuning paths to assess the same corpus across backbones. LFSBench evaluates long-form speech across scenarios and dimensions including content, acoustics, and expressiveness [17]. Russian LibriSpeech [18] releases utterance-level audiobook segments, and the multilingual YODAS crawl [19] includes unsegmented Russian YouTube audio but is not curated for single-speaker synthesis. To our knowledge, Balalaika-Longform is the first openly released Russian corpus curated for long-form TTS, and its all-attempt evaluation the first public long-form TTS evaluation for Russian. Our evaluation addresses a narrower training-data question: it controls input duration, retains failed attempts in the reported metrics, and evaluates whether a single generation reads the requested text.

3. CORPUS AND EVALUATION DESIGN

3.1. Corpus, data splits, and training arms

Balalaika-Longform v3.1 is released openly as WebDataset shards.¹ The open-source Balalaika pipeline [4] provided context-aware segmentation, multi-ASR ROVER transcription, and quality and speaker-purity screening. After loudness normalization, MossFormer2 speech enhancement [20] attenuated applause and background noise; the release and all training use this enhanced audio, and originals remain locatable through the stored source URL and time span.

The corpus contains 6,380 single-speaker Russian segments (189.13 h; 30–900 s) from 313 YouTube videos on 41 channels (seven hold 84% of the hours), mainly audiobook readings, lectures and talks, each carrying an uploader-declared Creative Commons Attribution flag, which is license evidence rather than a rights audit. Audio is 16-bit FLAC at the source rate (44.1 or 48 kHz, mostly stereo). Each segment carries a punctuated, capitalized GigaAM-v3-e2e-CTC [21] transcript; where available, the recognizer-output voting (ROVER) [22] consensus text over ASR hypotheses; one of 462 video-local diarization labels; and its source time span, URL and license evidence. Transcripts and speaker labels are automatic and uncorrected. Unit length is skewed: 3,538 units last 30–60 s, 1,733 last 1–2 min, 699 last 2–5 min, 276 last 5–10 min and 134 last 10–15 min, so the median is 46 s and the mean 107 s, yet units of at least 2 min hold 54% of the hours and the 410 units of at least 5 min hold 34%. We partition by YouTube channel before windowing: all segments of a channel belong to one split (Table 1); held-out units are longer on average (dev 255 s, test 169 s, train 102 s). All supervised fine-tuning (SFT) uses train only; dev supplies validation and the two-voice set’s texts, and dev and test channels, never used for gradient updates, supply reference voices. Channel identity is a conservative document/speaker proxy, supplemented by exact-text and 8-gram leakage checks and a speaker-embedding gate across splits.

Table 1. Channel-disjoint splits before training-text exclusions. The remaining 24 segments (0.88 h) come from seven held-out videos, one otherwise unrepresented, that fail the ASR-consistency gate; they are released separately and never used.

Split	Segments	Hours	Videos	Channels
Train	6,043	170.68	265	27
Dev	120	8.50	17	7
Test	193	9.07	30	7
Corpus	6,380	189.13	313	41

Training starts from the 6,043 released long units or 28,532 short windows that tile the same recordings (contiguous, at most 30 s, mean 21.5 s; 170.68 h, 1,257,944 words before per-view text exclusions); window boundaries and aligned transcripts are released as manifests with the code rather than as audio. For CosyVoice3, each long speech-token sequence concatenates its windows. Fine-tuning on the windows is Short-SFT and on the long units Long-SFT, both with ROVER text; Long-SFT-Punct keeps the Long-SFT audio tokens but substitutes punctuated, capitalized GigaAM-v3-e2e-CTC transcripts. This also changes words: 95.0% of the 6,041 training transcript pairs differ after evaluation normalization, with 5.4%

pooled word-edit disagreement, so this is not a punctuation-only ablation.

3.2. Fine-tuning to assess corpus utility

The main comparison changes training-sequence length while controlling exposure to the corpus. All CosyVoice3 arms start from the same checkpoint and fine-tune only the language model; the speech tokenizer, acoustic generator, and speaker encoder stay fixed. They use 3,000 updates with the same learning rate and loss normalization. Relative to Long-SFT, observed target-token exposure differs by 1.40% for Short-SFT and by 1.27% for Long-SFT-Punct.

We fine-tune Qwen3-TTS and VoxCPM2 with their public training code [23, 24], using punctuated text in both arms (their Short-SFT and Long-SFT-Punct). Within-backbone exposure differs by 0.17% for Qwen3-TTS and 0.30% for VoxCPM2. VoxCPM2’s training filter drops the 144 long units whose packed token sequences exceed its 8,192-position budget (517–899 s; 18.2% of the audio; retained units reach 661 s), while its short arm keeps windows from these recordings, so that comparison does not fully isolate duration from corpus composition. The CosyVoice3 and Qwen3-TTS arms share recording sets.

For F5-TTS, both arms fine-tune the full community Russian F5-TTS v1 Base checkpoint [25] with the flow-matching objective and punctuated transcripts. They traverse the same 6,038 recordings in the same order for three passes (18,114 updates), using intact recordings or short windows. Long/Short consume 510.92/510.26 hours of training audio, within 0.24% of the Qwen3-TTS arms’ audio budgets; final exponential-moving-average (EMA) weights are evaluated without score-based checkpoint selection. Both arms exclude records lacking usable windows; residual normalized transcript disagreement is 0.04%. Training, inference and evaluation code, the evaluation pairs with their 21 reference prompts, and per-attempt transcripts, alignment counts, stop reasons and window-level scores are released.²

3.3. State-continuous inference and evaluation sets

Continuous inference uses one target text and one voice-conditioning event, followed by an unsplit autoregressive trajectory or full-sequence flow solve. External splitting (including the CosyVoice3 text splitter), state resets, best-of- N selection and regeneration are disabled; all attempts are retained. Context, output and acoustic-buffer limits are identical across the arms of a backbone and documented with the code. They bind at about 1,200 words: VoxCPM2’s text fills about 4,900 of its 8,192 positions, so 16, 14 and 6 of 22 Base, Short-SFT and Long-SFT-Punct attempts end at that limit. F5-TTS estimates duration from prompt audio and text length; it has no end-of-sequence (EOS) decision, and planned duration is not a learned stopping measure.

The two-voice set (Table 2) combines six source texts (four development recordings and two constructed stress texts), five nested duration ranges and two held-out voices: 60 attempts per arm. The published-text set (Table 3) uses 50 pairs from 22 works and 21 reference speakers that contribute no fine-tuning data: eight LibriVox readers from Russian LibriSpeech [18] and 13 corpus speakers, six from test and seven from dev channels, first used in an earlier recorded-text pilot. Each work supplies one long passage near 1,200 normalized words, and 14 short and 14 medium prefixes target 75 and 300 words (ranges 60–90, 240–360 and 960–1,440 words, not reference-audio durations). Inputs are literal published spans

¹<https://huggingface.co/datasets/lab260/Balalaika-longform>

²<https://github.com/lab260ru/balalaika-longform>

Table 2. CosyVoice3 results on the earlier two-voice set (six texts, five duration buckets, two held-out voices; 60 attempts per arm) by input duration; all synthesis attempts are retained. Best value per row in bold.

Bucket	Target duration	Complete $\uparrow$ (%)				All-attempt WER $\downarrow$ (%)			
		Base	Short-SFT	Long-SFT	Long-SFT-Punct	Base	Short-SFT	Long-SFT	Long-SFT-Punct
B0	20–40 s	83.3	91.7	100.0	100.0	7.18	7.07	3.45	2.93
B1	60–90 s	58.3	41.7	83.3	75.0	14.56	17.04	11.70	9.61
B2	2–3 min	0.0	0.0	100.0	100.0	90.86	82.64	2.74	3.42
B3	4–6 min	0.0	0.0	41.7	75.0	98.07	97.77	47.22	36.66
B4	8–12 min	0.0	0.0	33.3	66.7	99.82	99.89	37.22	19.15
All	20 s–12 min	28.3	26.7	71.7	83.3	62.10	60.88	20.47	14.36

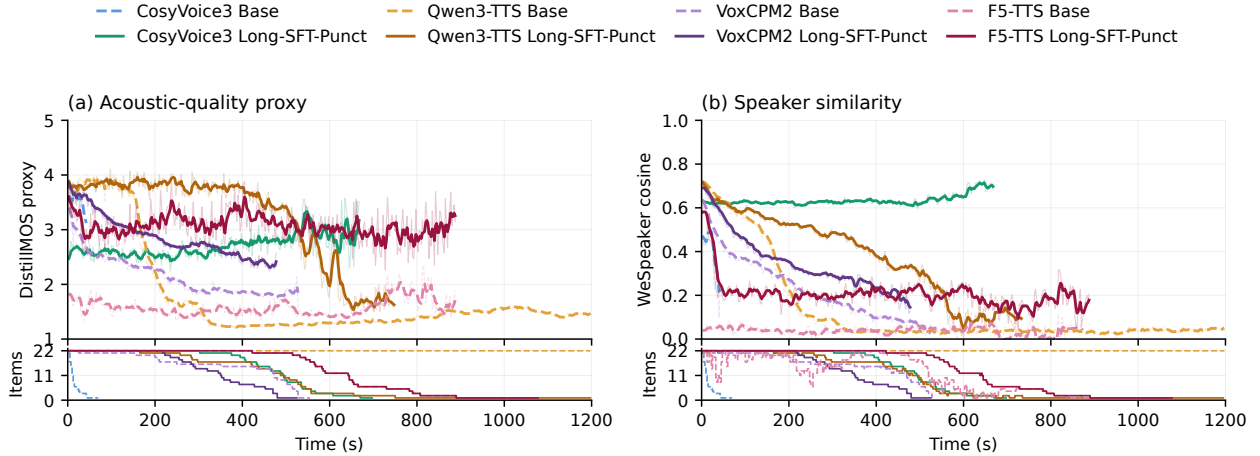


Fig. 1. Windowed diagnostics on the same 22 longest inputs per condition; Base and Long-SFT-Punct are shown, and all 13 conditions accompany the code release. Both metrics use 5-s windows with a 2.5-s hop; speaker windows require at least 1 s of voiced audio. Faint curves are item-weighted means, bold curves exponential smoothing ($\alpha = 0.3$); lower strips count contributing items (at least two per score).

screened against training and prior evaluation text; every arm uses the same pairs.

The 50 pairs are a deterministic, metadata-only subset of a planned 120-pair set (Section 5). All input is punctuated and capitalized. Among the CosyVoice3 arms, only Long-SFT-Punct matches this format in training; the two adapted arms of each other backbone share one training text format.

3.4. Evaluator and paired uncertainty

GigaAM-v3-RNNT [21] transcribes every generated output. Mean per-attempt WER counts substitutions, deletions and insertions relative to reference words; correct-word recall is aligned correct words divided by reference words. Unusable audio becomes an empty hypothesis. The published-text set has no paired human recording, so it uses no human-ASR floor or completion label.

Table 2 retains the earlier completion rule, frozen before SFT: an autoregressive EOS before 95% aligned text coverage is early; remaining outputs are labeled degraded when their WER is at least 30 points above a human-ASR floor or when they meet fixed repetition criteria. Watchdog stops retain their label. Constructed texts lack human recordings and skip the WER-floor test, so complete does not imply a faithful transcript.

For the published-text set, each backbone’s primary contrast compares its long and short arms (Long-SFT for CosyVoice3, Long-SFT-Punct for the other backbones) on the 22 longest inputs. Paired pointwise 95% intervals use 10,000 resamples of the 22 work clusters, retaining associated prefixes and voices; independently resam-

pling work and voice levels is a crossed-factor sensitivity check that widens but does not overturn any primary interval (CosyVoice3: $[-67.7, -34.6]$). Intervals condition on fitted checkpoints and inference seed and do not adjust for multiple comparisons. Intervals for the two-voice set instead resample its six source texts.

4. EXPERIMENTS

4.1. Earlier controlled duration comparison

Long-SFT improves continuous CosyVoice3 reading with two reference voices (Table 2). Short-SFT resembles Base, whereas Long-SFT gains 45.0 completion points [33.3, 56.7] over Short-SFT under matched exposure. Neither Base nor Short-SFT completes an item beyond 90 s. The sentence-split control completes all items at 3.84% WER; our continuous protocol claims no superiority over it. Repeated decoding shifts bucket percentages by up to ~ 20 points, so these single-seed rows are descriptive.

Punctuated transcripts add a secondary gain: Long-SFT-Punct reaches 83.3% completion, +11.7 completion points and -6.11 WER points over Long-SFT (Table 2), but this comparison changes both transcript source and formatting. Giving Long-SFT lowercased, unpunctuated input matched to its training format yields 88.3% completion and 10.07% WER, which Long-SFT-Punct (83.3%, 14.36%) does not improve on. The gain is therefore consistent with format matching; a separate benefit of punctuation content remains unestablished on this single-seed two-voice set. Three preregistered mixtures with external short clips did not improve Long-SFT and

Table 3. Published-text evaluation of all 13 continuous conditions by input length (14, 14 and 22 attempts per condition; texts from 22 works, 21 voices unseen in fine-tuning). All-attempt WER and correct-word recall include failed outputs; brackets are work-cluster bootstrap 95% intervals; text differences use unrounded means; best point estimate per backbone and column in bold (not a significance statement). Qwen3-TTS, VoxCPM2 and F5-TTS also use punctuated text for Short-SFT.

Backbone	Arm	All-attempt WER ↓ (%) [95% CI] by input length			Recall ↑ (%)
		~75 words	~300 words	~1,200 words	~1,200 words
CosyVoice3	Base	7.3 [4.1, 11.2]	84.8 [77.9, 90.9]	99.8 [99.7, 99.9]	0.2
	Short-SFT	7.7 [4.9, 10.8]	82.2 [72.5, 90.3]	99.9 [99.9, 100.0]	0.1
	Long-SFT	5.2 [2.8, 8.0]	13.8 [5.7, 24.1]	47.3 [37.2, 57.7]	69.5
	Long-SFT-Punct	3.6 [2.2, 5.0]	5.6 [2.7, 9.8]	16.6 [10.5, 23.2]	85.9
Qwen3-TTS	Base	3.9 [2.0, 6.1]	5.3 [2.5, 8.6]	66.9 [62.8, 70.8]	34.4
	Short-SFT	10.0 [7.1, 13.5]	49.6 [37.1, 61.3]	88.0 [85.5, 90.3]	12.2
	Long-SFT-Punct	8.3 [5.3, 11.7]	7.7 [5.6, 9.8]	35.2 [27.0, 43.9]	67.1
VoxCPM2	Base	12.4 [6.8, 18.7]	14.2 [3.4, 29.4]	94.9 [91.0, 98.0]	5.2
	Short-SFT	4.4 [2.2, 6.9]	4.1 [3.2, 4.9]	90.5 [86.9, 93.9]	10.5
	Long-SFT-Punct	3.4 [1.6, 5.7]	2.4 [1.8, 3.0]	49.0 [40.8, 56.7]	54.9
F5-TTS	Base	45.0 [34.7, 54.7]	98.7 [97.7, 99.6]	99.7 [99.7, 99.8]	0.3
	Short-SFT	44.2 [34.9, 52.8]	97.6 [96.9, 98.3]	99.3 [99.2, 99.5]	0.7
	Long-SFT-Punct	27.4 [20.9, 34.2]	81.4 [79.5, 83.2]	94.7 [94.4, 95.0]	5.4

increased repetition.

4.2. Content fidelity across backbones

The completed published-text comparison tests the training views on new source passages across four backbones (Table 3). For the three autoregressive models, Long-minus-Short WER changes at approximately 1,200 words, in percentage points (pp), are CosyVoice3: -52.7 pp [-62.7 , -42.2]; Qwen3-TTS: -52.7 pp [-61.7 , -43.4]; VoxCPM2: -41.5 pp [-47.7 , -35.4]. Negative changes indicate fewer word errors. All are paired effects on the same 22 work passages, not 50 independent attempts. The gap opens well before that length: at about 300 words CosyVoice3 Base and Short-SFT already reach 84.8% and 82.2% WER, while Long-SFT-Punct stays at 5.6%; at 1,200 words it improves on Base by 83.2 pp [-89.3 , -76.6]. Against Base rather than Short-SFT the changes are -52.5 , -31.6 and -46.0 pp, and at about 75 words Qwen3-TTS Long-SFT-Punct is 4.4 pp worse than Base [1.9, 7.2], so the benefit is specific to long inputs. The released stop logs show how arms fail at 1,200 words. Every CosyVoice3 Base and Short-SFT attempt emits EOS early (median 9.7 s and 5.3 s of audio), and Qwen3-TTS Short-SFT stops after a median 60 s, consistent with windows teaching a stopping decision near the window length; this is how Short-SFT can be worse than Base (Qwen3-TTS: 88.0% vs 66.9% WER).

4.3. Flow matching and inference protocol

F5-TTS extends the paired comparison to a flow-matching model. Its longest-input Long-minus-Short WER change is -4.6 pp [-4.9 , -4.4], but its absolute WER in Table 3 shows that a relative change alone does not establish reliable long-text reading. The duration schedule is shared across checkpoints, so this comparison evaluates content realization rather than an EOS decision. Its largest gain is on short inputs: at about 75 words Long-SFT-Punct lowers WER from 44.2% to 27.4%.

Chunking provides a distinct practical reference. On the *earlier* recorded-text 53-pair set, F5-TTS Base WER was 3.58% with sentence chunking and 73.35% with continuous synthesis. Those values use different texts from Table 3, are not rows of the published-text comparison, and establish no published-text ranking.

4.4. Evaluation checks and practical limits

The published-text WER uses one RNNT recognizer; checks with a second recognizer earlier supported the CosyVoice3 ranking on recorded texts. Figure 1 adds windowed DistillMOS [26] and We-Speaker [27] measurements on the published-text outputs. F5-TTS Long-SFT-Punct has higher DistillMOS than Base despite high WER; speaker similarity declines over time for Qwen3-TTS and VoxCPM2 but stays flat for CosyVoice3 (Fig. 1b). Curves condition on scorable windows, with support shown below. Truncated outputs can score well, so these proxies complement all-attempt WER and do not replace listening tests.

5. LIMITATIONS

The published-text set covers 22 literary works and 21 voices, and its 50 pairs were chosen from a planned 120 by a metadata-only rule after preliminary results had been seen. Each arm has one training seed, so intervals exclude retraining and decoding variation. GigaAM supplies training transcripts and scores outputs, and no listening study was run. Transcript-source, formatting and VoxCPM2 corpus-composition effects are not isolated, and one F5-TTS model does not represent all non-autoregressive systems. Corpus transcripts and speaker labels are automatic.

6. CONCLUSION

Balalaika-Longform preserves long continuous Russian speech with paired short-window training views. Across CosyVoice3, Qwen3-TTS, VoxCPM2 and F5-TTS, a two-voice study and a published-text comparison report absolute content fidelity and paired Short/Long changes; the intervals favor long targets for the three autoregressive backbones and by a small margin for F5-TTS, whose reading remains unreliable. The corpus supports studying sustained reading while keeping content fidelity and perceptual quality distinct.

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7. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using speech recordings that their uploaders published on YouTube with a Creative Commons Attribution declaration. No recordings were made for this work and no experiment involved human participants, so ethical approval was not required. The uploader’s declaration is license evidence; it is not an independent audit of rights in the underlying texts or performances and does not establish speakers’ consent to use in speech synthesis. The corpus is intended for research on long-form synthesis. Balalaika-Longform is distributed through <https://huggingface.co/datasets/lab260/Balalaika-longform> with per-source provenance and license evidence. Original recordings remain governed by their respective source licenses; the release does not relicense third-party material. Attribution corrections and removal requests are handled through the repository’s discussion page and the contact address given above.

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